

Lab 9

Neighborhoods

with data from the New York Times, Census,
& the Data and Democracy Lab

DATA 20195 Spring 2026

Emma Ross and Michael Xiao



The New York Times

An Extremely Detailed Map of New York City Neighborhoods

By **Larry Buchanan, Josh Katz, Rumsey Taylor and Eve Washington**
October 29, 2023

 Share full article  

Astoria

Long Island City

Williamsburg

Greenpoint

Blocks

Borders

Stats

Comments

An Extremely Detailed Guide to This Extremely Detailed Map



[Add your neighborhood >](#)

Let's explore for a few minutes...

 mapbox

 © Mapbox © OpenStreetMap [Improve this map](#)

NYT Data in the notebook

- 26,582 submissions, covering 379 neighborhood labels

	submission_id	neighborhood	geometry
0	s37680	Rego Park	POLYGON ((-74.266 40.77571, -74.26758 40.7758,...
1	s8242	Carroll Gardens	POLYGON ((-74.18548 40.71266, -74.1855 40.7126...
2	s19091	Mariners Harbor	POLYGON ((-74.14712 40.62561, -74.17969 40.632...
3	s12920	Graniteville	POLYGON ((-74.15735 40.61335, -74.15739 40.613...
4	s25108	Rossville	POLYGON ((-74.2175 40.54655, -74.21751 40.5465...
...	...	...	...
26577	s2538	Bellerose	POLYGON ((-73.70212 40.72902, -73.71837 40.726...
26578	s2521	Bellerose	POLYGON ((-73.71749 40.74395, -73.71748 40.743...
26579	s11276	Floral Park	POLYGON ((-73.71038 40.71526, -73.71954 40.715...
26580	s2533	Bellerose	POLYGON ((-73.71354 40.73622, -73.71353 40.736...
26581	s21056	Murray Hill (Queens)	POLYGON ((-73.70508 40.73165, -73.70542 40.729...

26582 rows x 3 columns

- Census blocks with weights per user label
- Of all the users that labelled *Block 360850323001007*, **51.5%** labelled it in **Mariners Harbor** and **48.5%** labelled it in **Graniteville**

```
list(block_weights.items())[:10] # Census block GEOID -> neighborhood weights
```

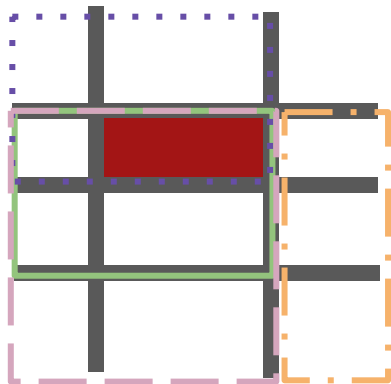
```
[('360850323001008', {'Mariners Harbor': 0.514, 'Graniteville': 0.486}),  
(('360850323001007', {'Mariners Harbor': 0.515, 'Graniteville': 0.485}),  
(('360850323001005',  
  {'Mariners Harbor': 0.393, 'Graniteville': 0.312, 'Elm Park': 0.295}),  
(('360850323001006',  
  {'Mariners Harbor': 0.361, 'Graniteville': 0.349, 'Elm Park': 0.29}),  
(('360850323001011',  
  {'Mariners Harbor': 0.35, 'Graniteville': 0.362, 'Elm Park': 0.288}),  
(('360850323001004', {'Mariners Harbor': 0.589, 'Elm Park': 0.411}),  
(('360850323001012',  
  {'Mariners Harbor': 0.35, 'Graniteville': 0.362, 'Elm Park': 0.288}),  
(('360850291021034', {'Bulls Head': 0.83, 'Graniteville': 0.17}),  
(('360850323001020',  
  {'Mariners Harbor': 0.319, 'Graniteville': 0.362, 'Elm Park': 0.319}),  
(('360850291022019', {'Travis': 1}))]
```

Neighborhood Cores: Label Consensus

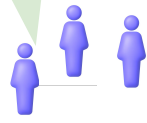
Among submissions that include **this block**, what share labeled it as Upper West Side?

1/2

“What do New Yorkers call **this block**?”



This is
Upper
West
Side



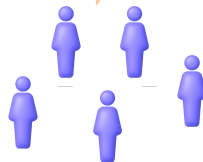
This is
Manhattan
Village



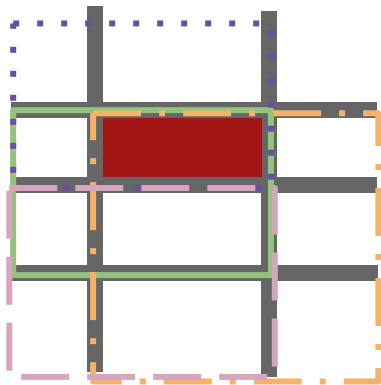
This is SOHO



This is SOHO



Neighborhood Cores: Block Inclusion Consensus



Among submissions about the Upper West Side, how many include **this block**?

4/6

This is
Upper
West
Side

This is
Upper West
Side

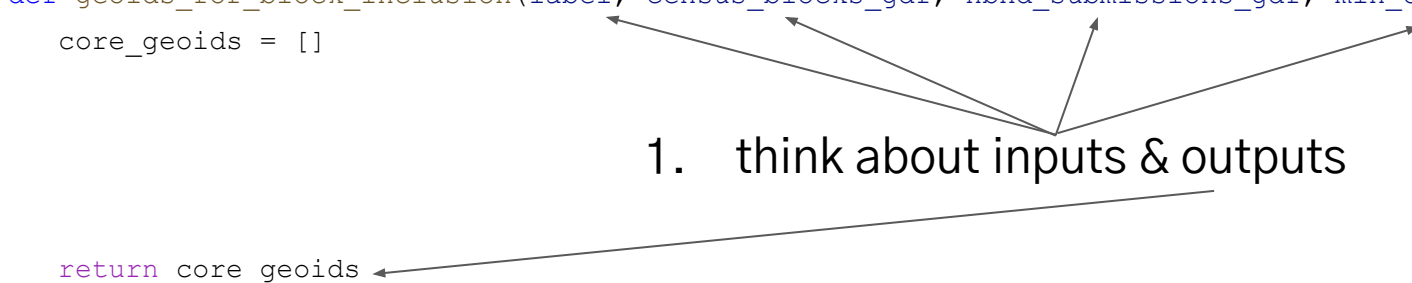
This is
Upper
West Side

This is SOHO

“How many New Yorkers say **this block** belongs to the ___ neighborhood?”

Task

```
# This function should return a list of block geoids where, among the people  
# that drew a polygon for the neighborhood with the label `label`, at least  
# `min_consensus` share of those submitted polygons intersect the block.  
def geoids_for_block_inclusion(label, census_blocks_gdf, nbhd_submissions_gdf, min_consensus=0):  
    core_geoids = []  
  
    1. think about inputs & outputs  
  
    return core_geoids
```



2. Look at the columns available in the gdfs, what will be of use to you?

(called `block_gdf`, `submissions_gdf` in notebook)

Which gdf should you iterate over?

Task

```
# This function should return a list of block geoids where, among the people
# that drew a polygon for the neighborhood with the label `label`, at least
# `min_consensus` share of those submitted polygons intersect the block.
def geoids_for_block_inclusion(label, census_blocks_gdf, nbhd_submissions_gdf, min_consensus=0):
    core_geoids = []
```

3. what other values or data tracking objects (i.e. a numpy array) should you initialize here?

```
#go through rows that have the neighborhood label of interest
for _, row in nbhd_submissions_gdf[nbhd_submissions_gdf["neighborhood"] == label].iterrows():
```

4. for each row, i.e. each ‘neighborhood’ entry – what do we need to do to see which blocks it intersects with?

```
return core_geoids
```

Task

```
# This function should return a list of block geoids where, among the people
# that drew a polygon for the neighborhood with the label `label`, at least
# `min_consensus` share of those submitted polygons intersect the block.
def geoids_for_block_inclusion(label, census_blocks_gdf, nbhd_submissions_gdf, min_consensus=0):
    core_geoids = []
    block_intersect_counts = np.zeros(len(census_blocks_gdf))
    overall_count = len(nbhd_submissions_gdf[nbhd_submissions_gdf["neighborhood"] == label])
    #go through rows that have the neighborhood label of interest
    for _, row in nbhd_submissions_gdf[nbhd_submissions_gdf["neighborhood"] == label].iterrows():
        #check which blocks the neighborhood intersects, adds 1 to np array at that block's index if so
        block_intersect_counts += census_blocks_gdf.intersects(row.geometry)
```

5. Now we have the counts for how often each block is included
block_intersect_counts , and we have the total number of entries for this
neighborhood label overall_count.

How do we pull these together with the min_consensus parameter
to define the core_geoids list?

```
return core_geoids
```


Task

```
def geoids_for_block_inclusion(label, census_blocks_gdf, nbhd_submissions_gdf, min_consensus=0):
    core_geoids = []
    block_intersect_counts = np.zeros(len(census_blocks_gdf))
    overall_count = len(nbhd_submissions_gdf[nbhd_submissions_gdf["neighborhood"] == label])
    #go through rows that have the neighborhood label of interest
    for _, row in nbhd_submissions_gdf[nbhd_submissions_gdf["neighborhood"] == label].iterrows():
        #check which blocks the neighborhood intersects, adds 1 to np array if so
        block_intersect_counts += census_blocks_gdf.intersects(row.geometry)

    #calculate the percentage / rate at which that block was included by all submitters
    percentage = block_intersect_counts/overall_count

    #select only blocks with percentages that meet the threshold
    for i in range(len(percentage)):
        if percentage[i] >= min_consensus:
            core_geoids.append(census_blocks_gdf.index[i])

    return core_geoids
```

Shannon Entropy

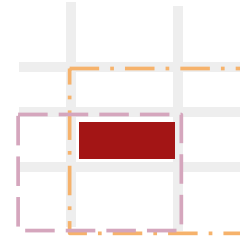
$$H = - \sum_{i=1}^n w_i \log(w_i)$$

Total entropy:
Measures the level
of controversy over
the neighborhood
label

Sum over all weights,
in our case –
sum over all
neighborhoods a
block was associated
with

$$0 < w_i < 1$$

$$-\infty < \log(w_i) < 0$$



This is
Upper
West
Side



This is SOHO



Upper West Side weight = $\frac{2}{5} = .4$
SOHO weight = $\frac{3}{5} = .6$

$$-(.4 \log_2(.4) + .6 \log_2(.6))$$

$$(-.4 * -1.3219) + (-.6 * -0.736)$$
$$H = 0.529 + 0.442 = 0.971$$

Shannon Entropy

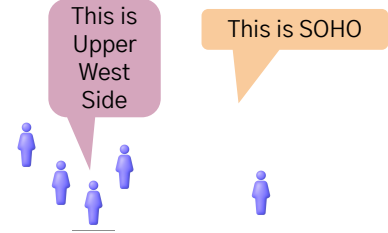
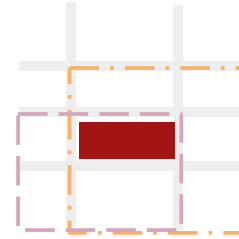
$$H = - \sum_{i=1}^n w_i \log(w_i)$$

↑
Total entropy:
Measures the level
of controversy over
the neighborhood
label

↑
Sum over all weights,
in our case –
sum over all
neighborhoods a
block was associated
with

↑
 $0 < w_i < 1$

↑
 $-\infty < \log(w_i) < 0$



Upper West Side weight = $4/5 = .8$
SOHO weight = $1/5 = .2$

$$-(.8 \log_2(.8) + .2 \log_2(.2))$$

More agreement: **lower** H, or Shannon entropy $H = 0.71$

Most agreement when **all weight in one neighborhood**: $1 * \log(1) = 0$

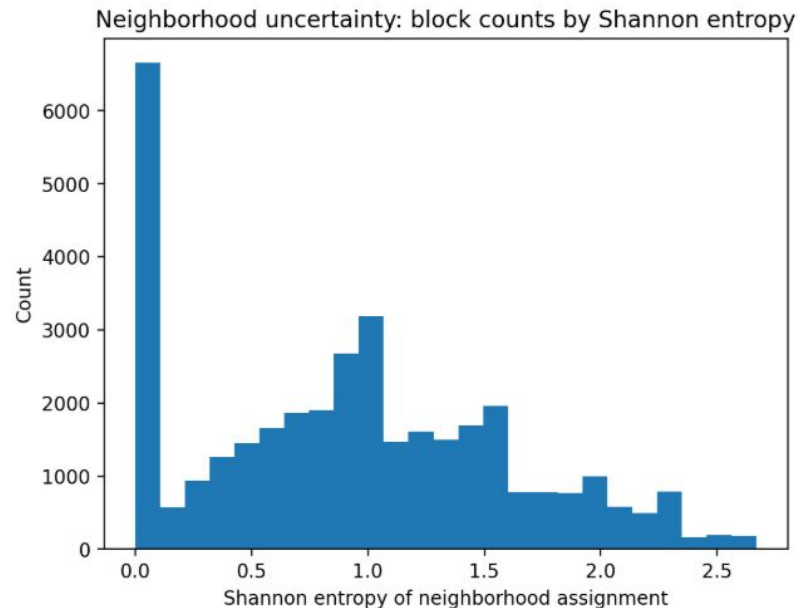
Least agreement when **weight divided equally across k neighborhoods**:

$$k * 1/k * \log(1/k) \longrightarrow \infty \text{ as } k \longrightarrow \infty$$

Shannon Entropy

$$H = - \sum_{i=1}^n w_i \log(w_i)$$

H : Total entropy: Measures the level of controversy over the neighborhood label
 $\sum_{i=1}^n w_i \log(w_i)$: Sum over all weights, in our case - sum over all neighborhoods a block was associated with
 $0 < w_i < 1$
 $-\infty < \log(w_i) < 0$



The Shannon entropy has a big peak at $H=0$ and a few other peaks, like one right near $H=1$. Where are the other peaks and why?

- $H=1$, 2 neighborhoods, weights .5
 $-2 * (.5 \log_2(.5)) = 2 * .5 * 1 = 1$
- $H=1.5$, 3 neighborhoods weights $1/3$
 $-3 * (1/3 \log_2(1/3)) = 3 * 1/3 * 1.5 \text{ ish} = 1.5 \text{ ish}$

Data & Democracy Lab Study: Participatory Mapping & Communities of Interest

Crowdsourced data on **communities of interest**.

Submissions for Michigan

districtr_id		submission_title	submission_text	area_name		area_text	blocks_2020	labels
0	17124-1	Dearborn Cultural District	Cultural venues in Dearborn. Important that th...	Community 1		None	[261635799004005, 261639838003006, 26163983800...	[]
1	17128-1	Dearborn neighborhoods near Ford Field	Area includes residents west of Brady, north o...	Dearborn Ford Field Neighbors	Neighborhoods using Ford Field, north of Michi...		[261635749001029, 261635749001005, 26163579900...	[]
2	17892-1	Example: Grand Rapids center and exurbs	This map shows the Grand Rapids urban center a...	urban center	The denser and more urbanized part of Grand Ra...		[260810016002007, 260810009002022, 26081013300...	[cities]
3	17892-2	Example: Grand Rapids center and exurbs	This map shows the Grand Rapids urban center a...	exurbs	Extending past Hudsonville to the West, these ...		[260810133002001, 260810112003002, 26081011200...	[cities]
4	17892-1	ADA reviews	0	urban center	The denser and more urbanized part of Grand Ra...		[260810016002007, 260810009002022, 26081013300...	[cities]

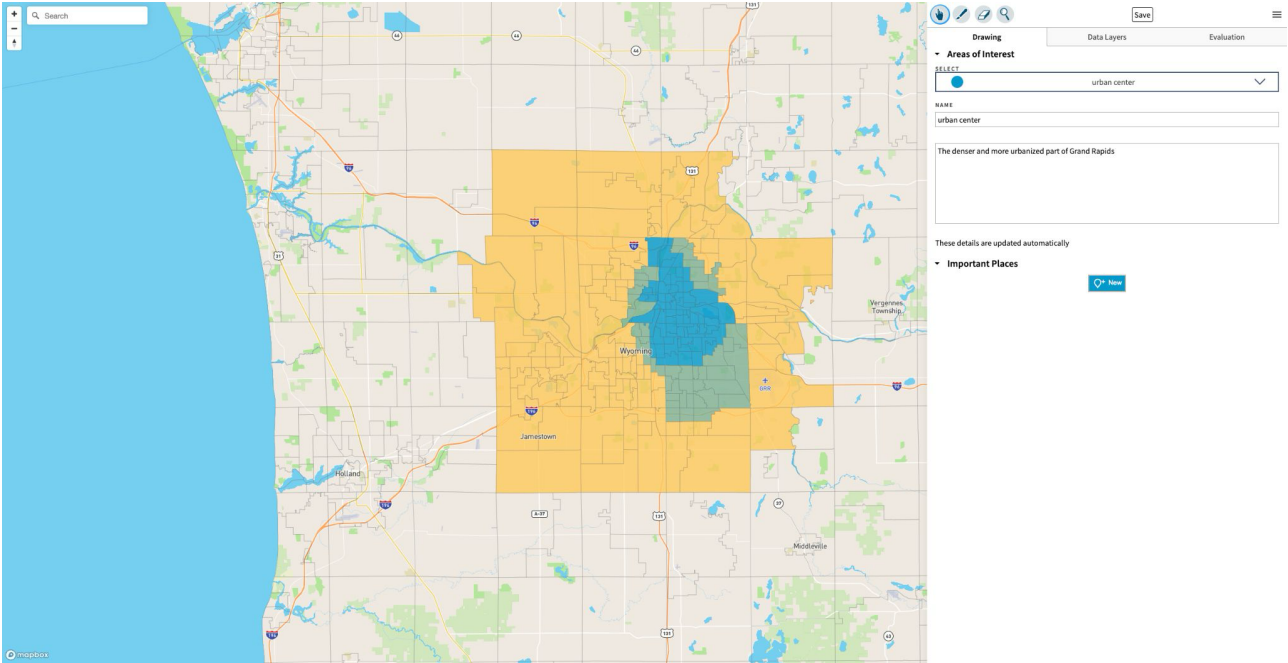
User-submitted information

Each row is an area a user submitted.

A submission includes 2 areas **here**.

districtr_id	submission_title	submission_text	area_name	area_text	blocks_2020	labels
0	17124-1	Dearborn Cultural District	Cultural venues in Dearborn. Important that th...	Community 1	None	[261635799004005, 261639838003006, 26163983800...
1	17128-1	Dearborn neighborhoods near Ford Field	Area includes residents west of Brady, north o...	Dearborn Ford Field Neighbors	Neighborhoods using Ford Field, north of Michi...	[261635749001029, 261635749001005, 26163579900...
2	17892-1	Example: Grand Rapids center and exurbs	This map shows the Grand Rapids urban center a...	urban center	The denser and more urbanized part of Grand Ra...	[260810016002007, 260810009002022, 26081013300...
3	17892-2	Example: Grand Rapids center and exurbs	This map shows the Grand Rapids urban center a...	exurbs	Extending past Hudsonville to the West, these ...	[260810133002001, 260810112003002, 26081011200...
4	17892-1	ADA reviews	0	urban center	The denser and more urbanized part of Grand Ra...	[260810016002007, 260810009002022, 26081013300...

<https://districtr.org/coi/17892>



A better process?

- crowdsourced data is super messy
- a lot of the Community of Interest (COI) submissions are not helpful for the organizations collecting the information

If *you* were running a community mapping project...

- How would you balance freedom of expression and setting limits to ensure you receive helpful, meaningful information?
- How might you communicate the necessary limits with community members?